WW-P-491b
May 1, 1967
SUPERSEDING
Fed. Spec. WW-P-491a
September 27, 1945

FEDERAL SPECIFICATION

PIPE-FITTINGS; CAST-IRON (THREADED): DRAINAGE

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers cast-iron drainage pipe fittings with threaded ends.

1.2 Classification.

1.2.1 Type. Cast-iron threaded drainage fittings shall be of the following types and styles (see 6.2):

Type I —Uncoated
Type II —Coated (Painted)
Type III—Zinc-coated (galvanized)
Style 1 —Regular threaded end.
Style 2 —Reduced size inlet threaded end.
Style 3 —Bushed threaded end.

1.2.2 Kinds, sizes, and dimensions. Cast-iron threaded drainage fittings shall be of the kinds, sizes, and dimensions specified (see 6.2), and shall be in accordance with USASI B16.12 and the applicable paragraph of this document, as indicated in table I.

Table I. Kinds, sizes, and dimensions

Fitting	Nominal pipe size (inches)	Dimensions	
		USASI B16.12	WW-P-491b
Elbows:			
90°; 90°-long; 90° 3-way; 45°	1¼ thru 8	table 4	—
90°; 45°	10	—	3.5.2
45° long; 60°; 22½°; 11¼°; 5⅝°	1¼ thru 6	table 4	—
90° reducer	1½ x 1¼ 2 x 1½	—	3.5.2
90° 3-way reducer	1¼ x 1¼ x 1½ 1½ x 1½ x 2	—	3.5.2
90° Street; 45° Street	1¼ thru 2	—	3.5.3
Tees:			
Straight tee	1¼ thru 6	table 5	—
Basin tee	1¼ thru 2	table 5	—
Reducing tee	1½ x 1½ x 1¼ thru 4 x 4 x 2	—	3.5.2
Basin cross	1¼ thru 2	table 5	—
90° Y-branches:			
90° short single; 90° long single	1¼ thru 6	table 5	—
90° short double; 90° double	1¼ thru 6	table 5	—

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Table I. Kinds, sizes, and dimensions (Continued)

Fitting	Nominal pipe size (inches)	Dimensions	
		USASI B16.12	WW-P-491b
90° Y-branches, reducing:			
90° short, single and double	1½ x 1½ x 1¼ thru 6 x 6 x 5	table 6	—
90° long, single and double	1½ x 1½ x 1¼ thru 6 x 6 x 5	table 6	—
45° Y-branches; single & double	1¼ thru 8	table 7	—
45° Y-branch, single	10	—	3.5.2
45° Y-branches: reducing, single & double	1½ x 1½ x 1¼ thru 8 x 8 x 6	table 8	—
Traps:			
P-trap	1¼ thru 4	table 13	—
Bath trap	1½; 2	table 13	—
Single vent running trap	1¼ thru 4	table 13	—
Double vent running trap	1¼ thru 4	table 13	—
Roof connections	1½ thru 4	table 12	—
Tucker connections	1 thru 6	table 12	—
Couplings	1¼ thru 6	table 9	—
Increases	2 x 1½ thru 6 x 5	table 10	—
Offsets	2 thru 6	table 11	—

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Standards:

Fed. Std. No. 123—Preservation, Packaging, and Packing Levels.
Fed. Std. No. 151—Metals, Test Methods.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles, and Auburn, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL-V-3—Valves, Fittings, and Flanges (Except for Systems Indicated Herein); Packaging of.

Military Standards:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-109—Inspection Terms and Definitions.
MIL-STD-129—Marking for Shipment and Storage.

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(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Publications:

A126—Gray Iron Castings for Valves, Flanges, and Pipe Fittings
A153—Zinc Coating (Hot-Dip) on Iron and Steel Hardware
A197—Cupola Malleable Iron

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, Pa. 19103.)

American Trucking Associations, Inc. (ATA)
National Motor Freight Classification Rules

(Application for copies should be addressed to American Trucking Associations, Inc., 1616 P Street, N.W., Washington, D.C. 20036.)

Uniform Classification Committee
Uniform Freight Classification Rules

(Application for copies should be addressed to Uniform Classification Committee, 202 Union Station, 516 W. Jackson Blvd., Chicago, Illinois 60606.)

United States of America Standards Institute (USASI)
B16.12–1965—Cast-Iron Threaded Drainage Fittings

(Application for copies should be addressed to United States of America Standards Institute, 10 East 40th St., New York, N.Y. 10016.)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

3.1 Material. Fittings shall be made from gray iron conforming to ASTM A126. The manufacturer may supply fittings made from malleable iron conforming to ASTM A197.

3.2 Coating.

3.2.1 Type I. Type I fittings shall be uncoated.

3.2.2 Type II. Type II fittings shall be painted with a thin, hard, adherent, rust-protective black lacquer or black enamel. All surfaces of the fittings, except threads, shall be coated.

3.2.3 Type III. Type III fittings shall be coated by the hot-dip process and so treated as to be thoroughly coated both inside and outside with a smooth, bright, adherent surface of zinc conforming to class A of ASTM A153. The coated fittings shall be free of blisters, dross, dirt, bare spots, rust, and coating defects. The fittings shall be coated before threading.

3.3 Threads. All fittings shall be threaded in accordance with USASI B16.12.

3.4 Dimensional tolerances. Tolerances on dimensions for all fittings, sizes 1¼ inch through 8 inches, shall be in accordance with USASI B16.12 and table II of this specification. Tolerances for the diameter and band width of 10-inch fittings shall be a plus 0.00-inch and a minus 0.010-inch. Tolerances for the center-to-end dimension of 10-inch fittings shall be ± 0.018 inch.

3.5 Dimensions.

3.5.1 Threaded ends for styles 1, 2, and 3. The female threaded ends of styles 1, 2, and 3 fittings shall conform to the dimensions specified in figure 1 and table II.

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Table II. Cast iron drainage fittings
Standard detail dimensions[1]

Nominal pipe size (inches)	Length of thread[1]	Length of thread chamber, face to shoulder[1]	Width of band[2]	Inside diameter of fitting[3]		Metal thickness, minimum[4]	Outside diameter of band[2]	Inside diameter of groove	
								Minimum	Maximum
A	B	C	E	F	F_1[5]	G	H	K	K
	Inches	Inches	Inches	Inches	Inches	Inches	Inches	Inches	Inches
1¼	0.42	0.71	0.71	1.38	1.25	0.18	2.39	1.66	1.73
1½	.42	.72	.72	1.61	1.50	.20	2.68	1.90	1.97
2	.44	.76	.76	2.07	2.00	.22	3.28	2.38	2.45
2½	.68	1.14	1.14	2.47	2.50	.24	3.86	2.88	2.98
3	.77	1.20	1.20	3.07	3.00	.26	4.62	3.50	3.60
4	.84	1.30	1.30	4.03	4.00	.31	5.79	4.50	4.60
5	.94	1.41	1.41	5.05	5.00	.38	7.05	5.56	5.66
6	.96	1.51	1.51	6.07	6.00	.43	8.28	6.62	6.72
8	1.06	1.71	1.71	7.98	8.00	.55	10.63	8.62	8.72
10	1.21	1.93	1.93	10.02	10.00	.69	13.12	10.75	10.85

* The values for "K" on style 3 fittings (see figure 1) correspond to reduced size inlet.

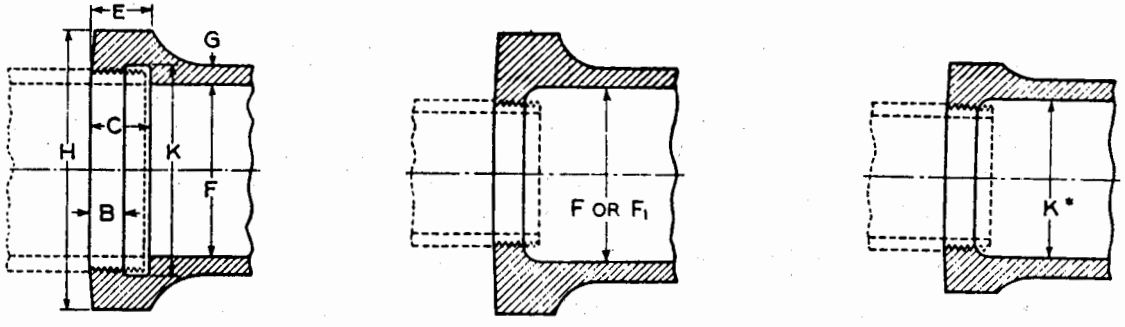
[1] Length of thread, B, and total length of thread chamber to shoulder, C, shall not vary from the dimensions shown in the table by more than plus or minus the equivalent of the pitch of one thread.

[2] The minus tolerances given in table I may be permitted on inspection to apply to the outside diameter of band, H, and the width of band, E.

[3] Inside diameters of fittings, F and F_1, shall not vary from dimensions given by more than ±1/32 in. for sizes 1¼ to 4 in. and ±1/16 in. for sizes 5 to 8 in.

[4] Patterns shall be designed to produce castings of a metal thickness given in the table. Metal thickness at no point shall be less than 90 percent of the thickness given in the table.

[5] The shoulder and groove are required on style 1 outlet connections. They are also required on inlet connections which are the same size as the outlet connections. Unless otherwise specified (see 1.2.1 and 6.2), inlet connections which are smaller than the outlet connections may be made as shown in styles 2 or 3 at the manufacturer's option.



Style 1
Regular

Style 2
Reduced size inlet

Style 3
Bushed

Figure 1. Threaded ends of cast iron drainage fittings.

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3.5.2 Elbows, tees, and Y-branches. The following size fittings are not manufactured to standard center-to-end dimensions. Threaded-end dimensions for these fittings shall conform to figure 1 and table II.

Fitting	Nominal pipe size (inches)
90-degree elbow (short)	10
45-degree elbow (short)	10
90-degree elbow reducer	1½ x 1¼
90-degree elbow reducer	2 x 1½
45-degree single Y-branch	10
Reducing tee	1½ x 1½ x 1¼
Reducing tee	2 x 2 x 1½
Reducing tee	2 x 2 x 1¼
Reducing tee	2½ x 2½ x 2
Reducing tee	3 x 3 x 1½
Reducing tee	4 x 4 x 3
Reducing tee	4 x 4 x 2

3.5.3 Street elbows; 90 degree and 45 degree. 90 degree and 45 degree street elbows shown in figures 2 and 3 are not made to standard center-to-end dimensions. Female threaded end dimensions shall conform to figure 1 and table II. The minimum length of male threads shall be as follows:

Nominal pipe size (inches)	Length of male thread (min. inches)
1¼	0.71
1½	.72
2	.76

3.6 Pressure test. When specified (see 6.2 and 6.4), fittings covered by this specification shall not leak when subjected to a minimum 60 p.s.i. air pressure in accordance with 4.4.1.

3.7 Identification marking. The manufacturer's name, initial, or other designation by which he can be easily identified, shall be legibly cast on each fitting.

3.8 Workmanship. The quality of workmanship shall be such as to produce fittings that are in accordance with the requirements of this specification. The fittings shall be free from sand and scale.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling for lot acceptance.

4.2.1 Inspection lot. For the purpose of sampling, a lot shall consist of all fittings of the same type, kind, and size offered for delivery at one time.

4.2.2 Sampling for examination. A random sample of fittings shall be selected from each lot in accordance with MIL-STD-105 at inspection level II. The acceptable quality levels (AQLs) shall be 1.5 percent defective for major defects and 4.0 percent defective for minor defects.

4.2.3 Sampling for tests.

4.2.3.1 Sampling for air pressure test. When specified (see 6.2), a random sample of fittings shall be selected from each lot offered to the Government in accordance with MIL-STD-105 at inspection level S-4. The acceptable quality level shall be 1.0 percent defective.

4.2.3.2 Sampling for chemical analysis. Unless otherwise specified (see 6.2), the manufacturer may submit a ladle analysis as proof that the fittings submitted meet the requirements of 3.1.

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4.2.3.2.1 When specified (see 6.2), two 2-ounce samples shall be taken from each lot and shall be subjected to chemical analysis specified in 4.4.2.

4.3 Examination. Each of the sample fittings selected in accordance with 4.2.2 shall be examined to verify compliance with this specification. Examination shall be conducted as specified in table III.

Table III. Classification of defects in accordance with MIL-STD-105

Categories	Defects
Major	
101	Kind of fitting not as specified.
102	Size not as specified.
103	Threads not as specified.
104	Surface coating not as specified.
105	Surface coating not uniform.
106	Cracks or blow holes present.
107	Dimensions exceeding tolerances.
Minor	
201	Identification marking not as specified.
202	Sand or scale on surface.

4.4 Test methods.

4.4.1 Air pressure. When specified (see 6.2 and 6.4), fittings selected shall be subjected to at least 60 p.s.i. air pressure while under water. Any leakage of air shall be cause for rejection.

4.4.2 Chemical composition. When specified (see 6.2), sample selected in accordance with 4.2.3.2.1 shall be analyzed in accordance with method 111.1 or method 112.1 of Fed. Std. 151.

4.5 Inspection of preparation for delivery. The preservation, packaging, packing, and marking of the fittings shall be inspected to determine conformance with the applicable requirements of section 5 of this specification.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A. The fittings shall be preserved and packaged in accordance with the applicable level A requirements of MIL-V-3.

5.1.2 Level C. The fittings shall be preserved and packaged in accordance with the supplier's standard practice.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. The fittings shall be packed in accordance with the applicable level A requirements of MIL-V-3.

5.2.2 Level B. The fittings shall be packed in accordance with the applicable level B requirements of MIL-V-3.

5.2.3 Level C. The complete equipment shall be packed in a manner which will insure arrival at destination in satisfactory condition and be acceptable to the carrier at lowest rates. Containers and packing shall comply with Uniform Freight Classification Rules, or with National Freight Classification Rules.

5.3 Marking (see 6.2).

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5.3.1 Civilian agencies. In addition to any special marking required by the contract or order, packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 Military agencies. In addition to any special marking required by the contract or order, packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Fittings covered by this specification are intended for general drainage applications.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.

(b) Type of finish required (see 1.2.1 and 3.2).

(c) Style of threaded end required (see 1.2.1 and 3.5.1).

(d) Kind and size of fitting required (see 1.2.2).

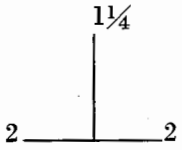
(e) When pressure test is required (see 3.6 and 6.4).

(f) When chemical analysis is required (see 4.2.3.2, 4.2.3.2.1 and 4.4.2).

(g) Selection of applicable level of preservation and packaging and level of packing required (see 5.1 and 5.2).

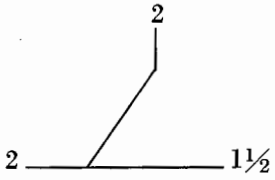
(h) Marking required (see 5.3).

6.3 In ordering reducing fittings, the kinds of fittings and sizes of the connecting ends should be specified. In giving the sizes of reducing tees and Y-branches, the larger run is given first, the smaller run next, and the inlet or branch last. The following diagram illustrates the above method:



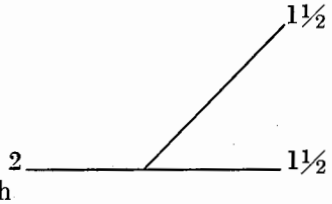
(2 x 2 x 1¼)

Reducing tee



(2 x 1½ x 2 reducing 90° short)

Y-branch



(2 x 1½ x 1½ reducing 45°)

Y-branch

6.4 Pressure test. Drainage fittings intended for low pressure service are not required to be pressure tested.

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Military Interest:

Custodians:
Army—ME
Navy—YD
Air Force—69

Review:
Army—ME
Navy—YD
Air Force—69

Preparing activity:
Navy—YD

Civilian Agencies Interest:
GSA–FSS
VA—DMS

Review and user information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current Federal Supply Classification Listing of DOD Standardization Documents.

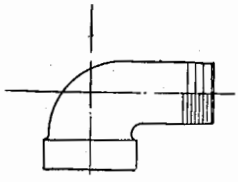


Fig. 2. 90 degree street elbow

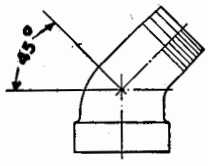


Fig. 3. 45 degree street elbow

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